

Direction generation of 3 MW peak-power picosecond pulses in a compact all-fiber all-PM CPA-free amplifier

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We report a 1030 nm all-polarization-maintaining (PM) all-fiber megawatt-peak-power picosecond amplifier without chirped pulse amplification (CPA). The compact fiber picosecond amplifier consists of a 0.8 m long highly Yb-doped photonic crystal fiber (PCF), a home-made 1030 nm all-fiber picosecond seed source, and a 60 W 976 nm pigtailed laser diode. The Yb-doped PCF is self-designed with a large core diameter of 41 μm , a low numerical aperture ($\text{NA} = 0.03$), and a high cladding absorption coefficient (25 dB/m) at 976 nm, offering the following merits: 1) only requiring short-length PCF as a gain medium and weakening nonlinear effects in favor of CPA-free of picosecond pulses, and 2) well maintaining excellent beam quality. The amplifier can deliver 21.7 μJ , 7.2 ps pulses at 505.8 kHz (corresponding to 3 MW peak power), and the maximum average power of 26 W@5.5 MHz with a 55.7% optical-to-optical conversion efficiency. Moreover, it operates at a near-diffraction-limited beam quality with $M_x^2 = 1.06$ and $M_y^2 = 1.1$. To the best of our knowledge, this is the highest peak power obtained from such a compact CPA-free all-PM all-fiber picosecond amplification system.

Keywords: fiber laser amplifier; picosecond pulse; photonics crystal fiber.

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1. Introduction

Picosecond lasers at 1 μm wavelength have been widely used in industrial processing^[1,2], such as laser scribing^[3], laser ablation^[4], laser drilling^[5], and laser stealth dicing^[6]. These precision machining processes usually require the laser parameters of <10 ps duration, MW peak power, pulse energy of several tens of μJ ^[7,8], and excellent beam quality ($M^2 < 1.2$). Currently, the primary technologies for generating large-energy, high-peak-power picosecond pulses include multi-stage solid-state crystal amplification^[9–12] and chirped pulse amplification (CPA)^[13–16], but suffer from the bulk optical components and complex spatial optical-path alignment, which degrades the compactness and robustness of picosecond amplification systems. Therefore, there are always strong motivations to explore new solutions for compact high-peak-power picosecond amplifiers.

A fiber-based master-oscillator power amplifier (MOPA) could be an ideal solution to directly amplify picosecond pulses,

due to the advantages of compactness, good beam quality, maintenance-free, and low cost^[17–19]. However, the peak power of picosecond pulses in conventional fiber MOPAs (e.g., step-index Yb-doped double-clad fiber) is typically limited to <0.5 MW^[20–22], because the strong optical nonlinearity happens or the optical beam quality deteriorates seriously^[23,24]. In recent years, the rod-type Yb-doped photonics crystal fiber (PCF) and the tapered Yb-doped fiber have been interestingly developed to boost the peak power of picosecond pulses above 1 MW^[25–27]. In 2013, Zhao *et al.* reported a rod-type PCF MOPA to yield 1 MW peak power, 3 μJ , and 3 ps pulses at 1 μm ^[28]. A record peak power of 4.5 MW directly from a 100 μm huge-core rod-type PCF MOPA was obtained in Ref. [29]. Moreover, a picosecond Yb-doped tapered fiber amplification system allowed the peak power as high as 1.2 MW at 1038 nm^[30]. However, because the rod-type PCF or tapered fiber is often incompatible for fusion splicing with standard fibers, such fiber-based MOPA systems still require free-space optical paths and bulk elements, sacrificing the all-fiber structure, the system robustness, and reliability.

Therefore, it is necessary to develop a compact, high-peak-power all-fiber picosecond MOPA system by exploiting a new type of fiber.

In this Letter, we design a new type of highly Yb-doped PCF to realize a 1030 nm MW-peak-power picosecond MOPA system in an all-PM all-fiber structure. The picosecond seed source is home-made by a figure-9 mode-locked fiber oscillator and pre-amplifier. A 41/145 μm highly Yb-doped PCF with a low numerical aperture (NA, ~ 0.03) and a high cladding absorption coefficient (25 dB/m) is used as the gain fiber of the main amplifier. We achieved a highest peak power of 3 MW (i.e., 21.7 μJ pulse energy and 7.2 ps duration) and a beam quality factor of $M_x^2 = 1.06$, $M_y^2 = 1.1$. Furthermore, the average amplification power can reach 26 W at 5.55 MHz with optical-to-optical conversion efficiencies as high as 55.7%.

2. Experimental Setup

The schematic diagram of the CPA-free high-peak-power picosecond laser amplification system based on a highly Yb-doped PCF is shown in Fig. 1(a). Our seed source is a 1030 nm all-PM all-fiber figure-9 mode-locked oscillator with pre-amplification, and the output pulse duration is 10.1 ps. The pulse repetition frequency is tunable from 8.33 MHz to 101 kHz by the pulse picker (CETC, SGGT200- 2D1). A 60 W 976 nm semiconductor laser diode serves as the pump source, which is injected into a 0.8 m highly Yb-doped PCF through a 976/1030 nm (6 + 1) $\times$ 1 pump-signal combiner. At this pump power, the amplifier can be thermally managed by passive cooling. To ensure the amplified signal's output power purity and optimize optical transmission efficiency, a cladding power stripper (CPS) and an end cap are integrated at the tail of the fiber. Importantly, all the pigtailed of the system are PM fibers.

The key component of the picosecond laser amplification system is the highly Yb-doped PCF, depicted in Fig. 1(b). Cross-sectional scanning electron microscope characterization

[Fig. 1(b)] reveals the fiber's geometric parameters; the core/cladding diameters of the highly Yb-doped PCF fiber are 41 and 145 μm , respectively, with a fiber core NA of about ~ 0.03 . The fiber incorporates symmetrically arranged panda-type stress rods (visible as two symmetrical elliptical regions) that induce birefringence, thus providing polarization-maintaining capabilities. This design facilitates robust polarization stability and allows for low-loss fusion splicing via precise alignment of the polarization axes. The strategically designed air-hole microstructure also achieves an enlarged mode field diameter of 34 μm , effectively reducing the intra-core power density to suppress nonlinear optical effects. Notably, this specialty fiber exhibits exceptional pump absorption characteristics, with a 976 nm pump absorption coefficient of up to 25 dB/m—significantly higher than that of conventional commercial counterparts [Fig. 1(c)]. Based on this, the amplifier can be designed with a short length, which helps reduce nonlinear phase accumulation. This outstanding absorption efficiency can effectively overcome the conventional energy limitations of all-fiber laser architectures, thereby enabling the generation of high-pulse-energy lasers.

In this experiment, the pulse trains and the radio frequency (RF) spectrum were measured by a 15 GHz photo-detector together with a digital oscilloscope (Agilent 54832D) and an electrical spectrum analyzer (Keysight E4440A), respectively. Besides, the accurate pulse duration and output optical spectrum were monitored by an optical spectral analyzer (OSA, ANDO AQ6315E) and an autocorrelator (Femtochrome FR 103-XL). The power measurement devices are all from Thorlabs (PM100D and S142C/S425C-L).

3. Results and Discussion

The seed source consists of a 1030 nm all-PM all-fiber mode-locked figure-9 oscillator with a fundamental frequency of 16.67 MHz and a three-stage fiber amplifier with Yb-doped single/double-clad fibers. Using a pulse picker, the repetition frequency can be tuned from 8.33 MHz to 101 kHz. Figure 2 presents the output characteristics of the all-PM all-fiber seed laser at a repetition rate of 505.8 kHz. As depicted in Fig. 2(a), the seed source exhibits a pulse duration of 10.1 ps. Besides, at 505.8 kHz, it exhibits a center wavelength of 1029.85 nm, a 3 dB bandwidth of 0.59 nm [as shown in Fig. 2(b)], and a pulse energy of 0.6 μJ . Meanwhile, optical spectrum characteristics and pulse energy of the seed source at other repetition frequencies are summarized in Table 1.

Based on the seed source, we achieved a maximum output power of 11 W, corresponding to the pulse energy of 21.7 μJ , and the peak power of 3.01 MW at the repetition frequency of 505.8 kHz, as shown in Fig. 3(a). Figure 3(b) shows the optical spectrum of the amplification system at the highest peak power, which exhibits a center wavelength of 1029.7 nm and a 3 dB spectral bandwidth of 3.18 nm. As shown in Fig. 3(c), a distinct pedestal is observed on the autocorrelation trace, and the pulse width is narrowed to 7.2 ps, which is caused by nonlinear effects.

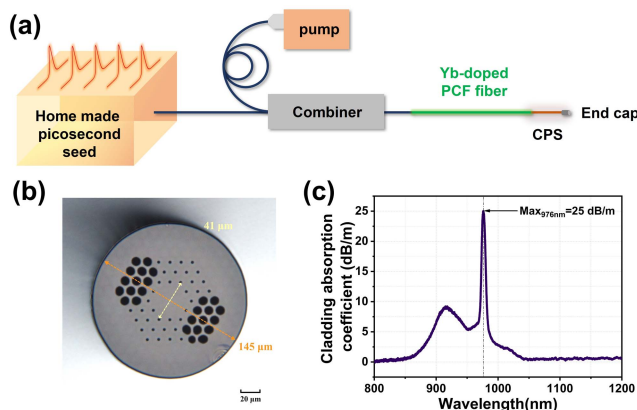


Fig. 1. (a) Experimental setup diagram of the CPA-free high-peak-power picosecond amplification system based on highly Yb-doped PCF. (b) Cross-sectional scanning electron microscope image of the highly Yb-doped PCF. (c) Absorption spectrum of the highly Yb-doped PCF.

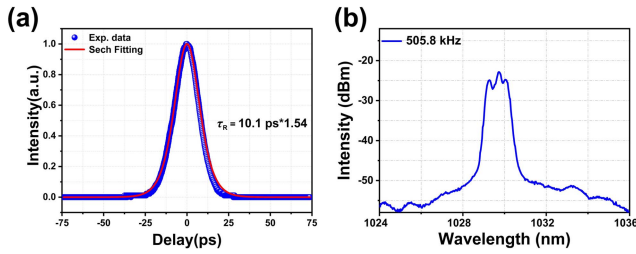


Fig. 2. Seed source at the repetition frequency of 505.8 kHz. (a) Autocorrelation trace. (b) Optical spectrum.

Table 1. Optical Spectrum Characteristics and Output Pulse Energies at All Repetition Frequencies.

Repetition frequency (kHz)	Center wavelength (nm)	3 dB bandwidth (nm)	Pulse energy (μ J)	Peak power (MW)
101	1029.69	2.16	1.72	0.17
150	1029.71	1.77	1.52	0.15
303	1029.69	1.17	0.85	0.08
505.8	1029.85	0.59	0.6	0.059
794	1029.67	0.78	0.4	0.039
1.1×10^3	1029.67	0.71	0.466	0.046
5.5×10^3	1029.65	0.39	0.101	0.01

The beam quality of the final output is shown in Fig. 3(d); the measured M^2 factors of the amplification system were $M_x^2 = 1.06$ and $M_y^2 = 1.1$, indicating a beam quality close to

that of an ideal Gaussian beam. In addition, the radio frequency (RF) spectrum as shown in Fig. 3(e) exhibits a single fundamental peak at 505.8 kHz with no observable spurious frequency components, and the signal-to-noise ratio (SNR) is ~ 60 dB, demonstrating excellent signal purity. Figure 3(f) characterizes the polarization degree of the linearly polarized light after the amplifier. The calculated polarization extinction ratio (PER) is 22 dB, indicating that the dominant polarization direction of the laser is clear. These results demonstrate that our self-made highly Yb-doped PCF is very suitable for the 10-ps CPA-free all-PM all-fiber high-peak-power amplification system.

To characterize the stability of the high-peak-power picosecond amplification system, we recorded the output power and optical spectrum over 30 min under an output power of ~ 10 W. As shown in Fig. 4(a), the pulse energy stability of the picosecond amplification system has a root mean square error (RMSE) of 0.09 W and an intensity fluctuation of 4.5%. In addition, Fig. 4(b) illustrates that the picosecond amplification system's central wavelength and spectral bandwidth are unchanged, indicating the long-term stability of the picosecond amplification system, and its suitability for high-precision industrial applications.

In order to explore the relationship between output power and repetition frequency, we systematically investigated the picosecond amplification system characteristics under different repetition frequencies. Using a pulse picker, we switched the repetition frequency to 101, 150, 303, 794, 1100, and 5500 kHz; the typical pulse trains are shown in Fig. 5(a). Furthermore, optical spectrum characteristics at different repetition frequencies are presented in Fig. 5(b). When the repetition frequency of the amplification system decreases from 5.5 MHz to 150 kHz, the peak power inside the fiber will increase, leading to a gradual enhancement of nonlinear effects within the PCF

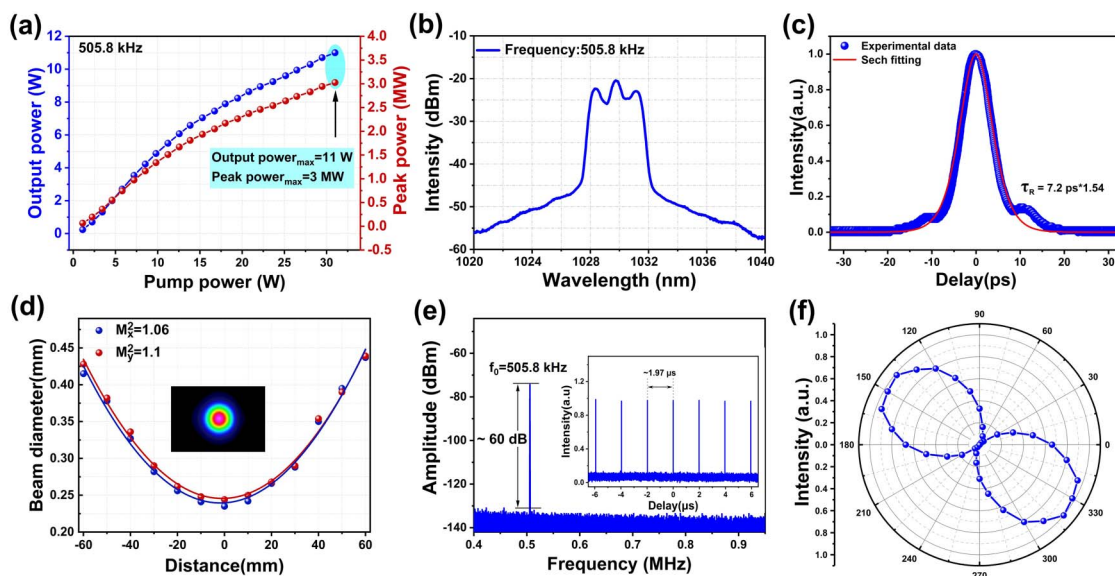


Fig. 3. Picosecond amplification system at the repetition frequency of 505.8 kHz. (a) Output power and peak power. (b) Optical spectrum. (c) Autocorrelation trace. (d) Output beam profile and beam quality. (e) RF spectrum; inset: pulse train. (f) PER distribution of the laser output.

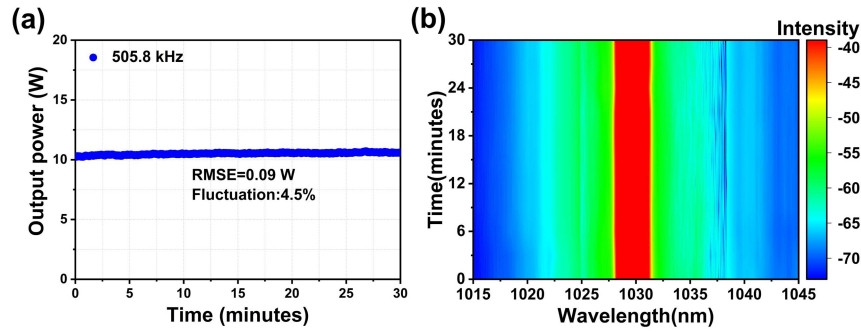


Fig. 4. 30 min stability of the picosecond amplification system. (a) Output power. (b) Optical spectrum.

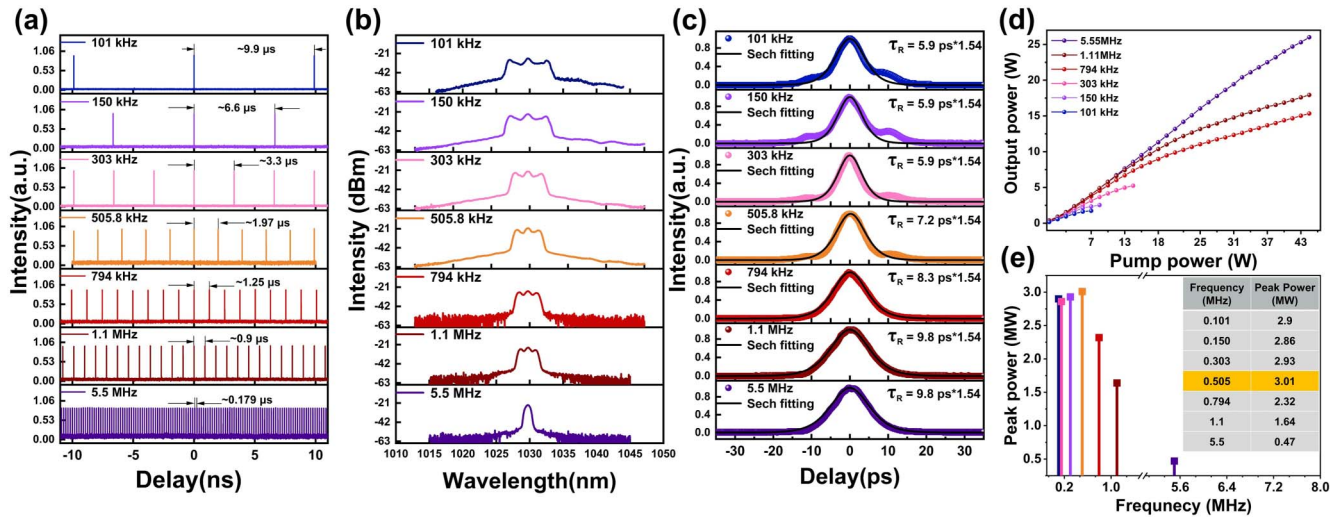


Fig. 5. Picosecond amplification system at different repetition frequencies. (a) Pulse train. (b) Optical spectrum. (c) Autocorrelation trace. (d) Output power. (e) Peak power.

and consequently a broadening of the optical spectrum. Meanwhile, the nonlinear effect is also manifested in the autocorrelation trace shown in Fig. 5(c). At 5.5 MHz, a 3 dB spectral width of 0.75 nm is obtained. Meanwhile, an average power of 26 W is achieved, as shown in Fig. 5(c), indicating that the amplifier has achieved an impressive optical-to-optical conversion efficiency of 55.7%. This further demonstrates the significant potential of our self-made, highly Yb-doped PCF in achieving high peak power and a robust performance of the picosecond amplification system.

4. Conclusion

In conclusion, we have demonstrated a high-peak-power, all-PM all-fiber CPA-free picosecond MOPA system at 1030 nm. The amplifier could directly emit 21.7 μ J, 7.2 ps pulses with a peak power as high as 3 MW. Moreover, the beam quality maintained the near-diffraction-limited factor of 1.06/1.11, and the polarization extinction ratio was as high as 22 dB. Such excellent results are attributed to the ability of this highly Yb-doped PCF to be precisely and low-loss fusion-spliced. In addition, we

systematically studied the output performance of the picosecond amplifier as the repetition rate, and the amplifier showed the 55.7% optical-to-optical conversion efficiency and the 26 W average power at 5.5 MHz. Such an amplifier possesses the notable advantages of compact all-fiber configuration, high peak power, and excellent beam quality, and could be promising for applications to precision processing in industry.

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